

INFRATHRONE (COREOPS TRACK)

WEEK 1 ASSIGNMENT PACK (Linux + Kubernetes Fundamentals)

Theme: Building the Foundation of Outage Handling

This assignment is designed to push you slightly outside your comfort zone while keeping everything achievable.

You must submit screenshots, notes, and your debugging explanation for each task.

The goal is not perfection; the goal is understanding.



Linux System Deep Dive (Mandatory)

Objective:

Understand how Linux behaves under stress so you can identify symptoms during an outage.

Tasks:

1. Run top for 2 minutes and record:
 - Highest CPU process
 - Highest memory process
 - Load average values
2. Trigger a CPU spike using: **yes > /dev/null &**

Now observe:

- Updated load average
 - Which CPU core is saturated
 - Process state
 - How the system behaves
3. Trigger a memory spike using this Python script:

```
a = []  
while True:  
    a.append(' ' * 10**7)
```

Observe:

- Free memory dropping
- Swap usage

- Any kernel warnings (dmesg | tail)
4. Kill the process using **signals**:
- **kill -SIGTERM <pid>**
 - **kill -SIGKILL <pid>**

Capture difference in behavior.

Submission:

- Screenshot of top before & after
- Explanation of what changed and why
- In 4–5 lines: *“How will I use this knowledge during a Kubernetes outage?”*

